Tuning the Step: How Wolfe Parameters Shape Optimization Performance

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Research Question:
How sensitive is algorithm performance to the choice of Wolfe line search parameters, and what are, the best choices for the parameters on average?

Introduction

The performance of optimization algorithms using Wolfe line search depends critically on the choice of parameters **(c1,c2)**, yet their practical impact is not well understood.

In this study, we investigate the sensitivity of algorithm performance to these parameters and identify choices that perform well on average. We focus on four representative methods–Gradient Descent, Newton, BFGS, and DFP–to examine how parameter selection interacts with different search directions.

Wolfe Line Search Conditions

$$f(x_k + \alpha_k d_k) \le f(x_k) + c_1 \alpha_k \nabla f(x_k)^T d_k$$
$$f(x_k + \alpha_k d_k)^T d_k \ge c_2 \nabla f(x_k)^T d_k$$

Algorithms

- GD
- Newton
- BFGS
- DFP

Range of Value for (c1,c2)

c1 $0 < c_1 < c_2 < 1$ **c2**

- controls acceptance condition
- too large → overly strict
- **(1e-5, 0.05)**

- controls aggressiveness
- too small → overly conservative
- **(0.1,0.99)**

Average Performance

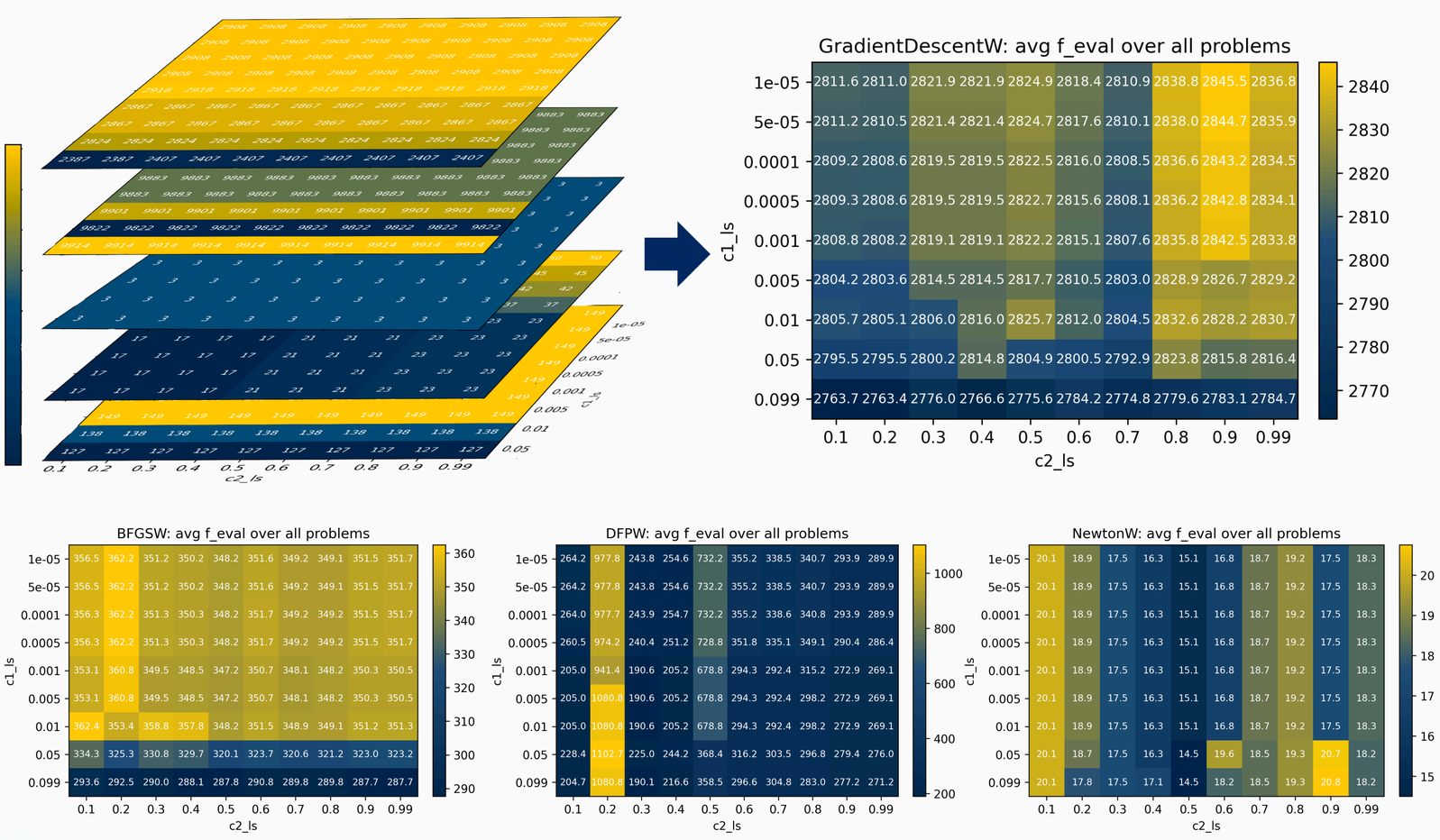

Figure **1**: Average performance **(**iteration amount**)** of each algorithm among different problems.

Performance Profiles

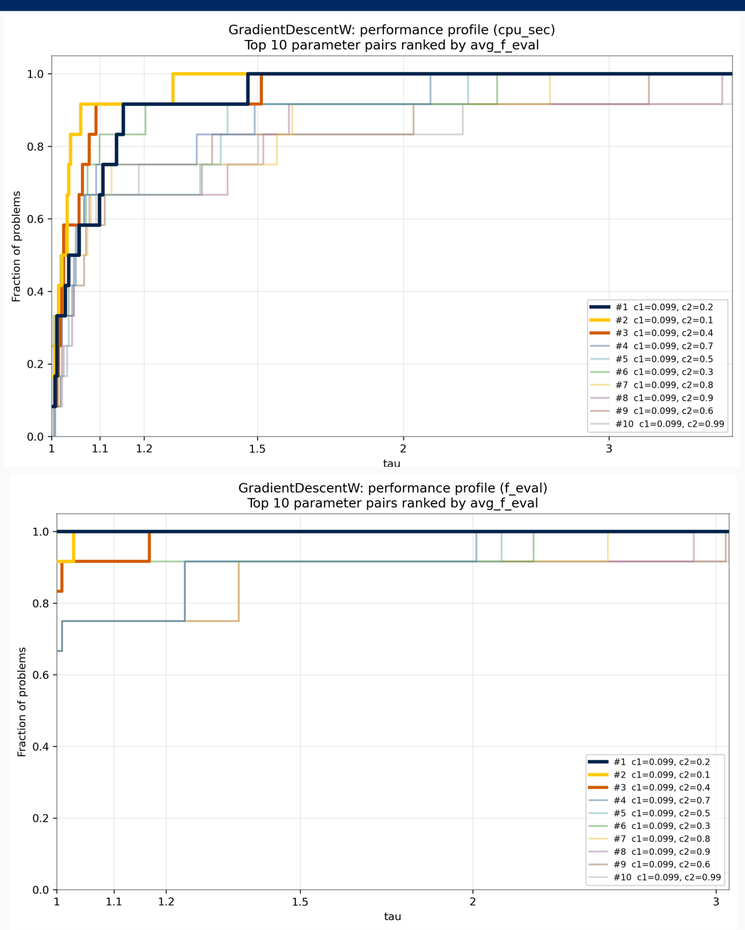

Figure **2**: Example: Although **(0.099,0.2)** for GD method shows slightly better outcome than **(0.099,0.1)**, its CPU running time is distinctly longer. Overall, we should choose **(0.099,0.1)**.

Sensitivity Analysis

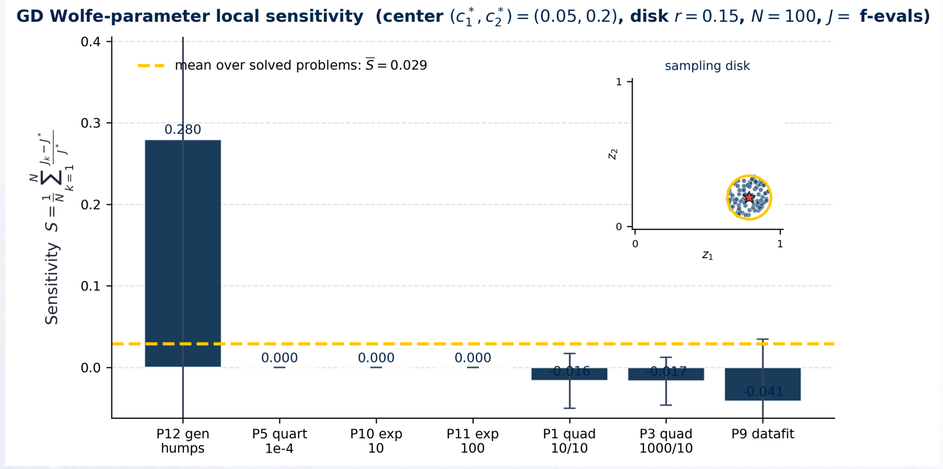

Figure **3**: Around **(c1*,c2*)=(0.05,0.20)**, in the normalized parameter space Nr, we uniformly sample N=**100** points and use J = f_eval as the performance metric.

$$S_p = \frac{1}{N}\sum_{k=1}^{N} \frac{J_k - J_p^*}{J_p^*} \qquad z_1 = \frac{\log_{10}(c_1) - \log_{10}(c_{1,\min})}{\log_{10}(c_{1,\max}) - \log_{10}(c_{1,\min})}, \qquad z_2 = \frac{c_2 - c_{2,\min}}{c_{2,\max} - c_{2,\min}}$$
$$N_r = \left\{(u,v) : (z_1(u) - z_1^*)^2 + (z_2(v) - z_2^*)^2 \le r^2\right\}$$

J_k is measured at the k-th sample in N_r, and J_p^* is measured at the center.

Average vs. Performance Profiles

- Average performance summarizes overall efficiency across problems, but it is sensitive to outliers and may favors parameter choices that perform extremely well on a small subset of cases **(**Fig. **2)**.
- Performance profiles emphasize robustness by measuring how consistently a method performs near the best across a wide range of problems.

Best Choice

In our study, we observe that the parameter setting with the best average performance does not always achieve the best performance profile. To identify reliable parameter choices, we adopt a combined criterion:

- First step – Average performance:
  select top-performing parameter pairs to filter out inefficient choices
- Second step – Performance profile:
  compare shortlisted candidates and select those with most robust performance

Method	(c1,c2)
GD	~~(0.099,0.2)~~ (0.099,0.1)
Newton	(0.05,0.5)
BFGS	~~(0.099,0.99)~~ (0.099,0.9)
DFP	(0.001,0.5)

Conclusion

Wolfe parameters matter, but their impact is uneven. To evaluate global robustness, we listed performance profiles of the top **10** parameter pairs, identifying a small c**1** and a large c**2** as the best average choices. Complementing this, our local sensitivity analysis shows that around GD's optimum, a **15%** perturbation in the normalized log c_**1**, c_**2** space changes f-evals by under **5%** on **6/7** convergent problems. The lone outlier is the non-convex genhumps problem. Ultimately, GD is insensitive where it works and sensitive where it struggles–precise tuning is primarily needed for hard, non-convex landscapes.